PROJECT REPORT

NETWORK REQUEST MANAGEMENT

ServiceNow Service Catalog and Network Request Automation

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Student Name: Bantupalli Niharika

Project Documentation

Prepared based on the implemented project and reference report structure.

Technology: ServiceNow
Application: Network Request Management
Project: Automated Network Request & Fulfillment Management System

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1. INTRODUCTION

1.1 Project Overview

The **Automated Network Request & Fulfillment Management System** is a ServiceNow-based solution designed to simplify and automate the process of requesting network-related services.

Employees can submit network requests through a centralized **Service Catalog**. The system captures required information such as device type, connection requirements, and request details. Catalog UI Policies and validations ensure that users provide relevant and complete information.

After submission, ServiceNow creates the request records and processes the request through an approval workflow. Once approved, a Network Task is created and assigned to the Network Fulfillment Team for processing.

High-Level Flow

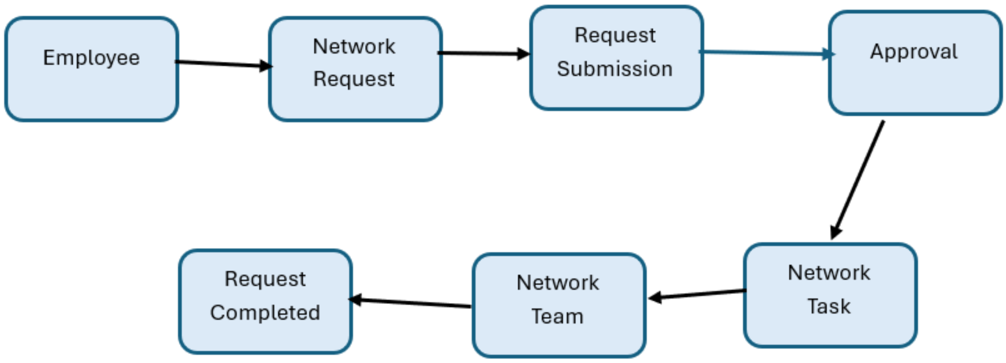


Figure 1: High-level workflow of the Automated Network Request Management System.

1.2 Purpose

The purpose of this project is to provide a centralized and structured process for managing network service requests.

The system aims to:

- Simplify network request submission.
- Capture complete request information.
- Reduce manual processing.
- Automate approvals.
- Automatically create fulfillment tasks.
- Assign tasks to the Network Team.
- Improve request tracking and visibility.

2. IDEATION PHASE

2.1 Problem Statement

Employees may face difficulties when requesting network-related services because requests can be submitted through unclear or unstructured processes. Incomplete information and manual processing can cause delays in approvals and fulfillment.

Problem Statement Canvas

Parameter	Description
User	Employees requesting network services
Problem	Network requests may be unclear or incomplete
Need	A simple and structured request process
Impact	Delays in approval and fulfillment
How Might We?	How might we provide a centralized system for submitting and managing network requests efficiently?

2.2 Empathy Map Canvas

Employee / End User Perspective

Dimension	Employee Perspective
Says	I need a simple way to request network services.
Thinks	I want a clear form with relevant options.
Does	Opens the portal, selects Network Request, and submits details.
Feels	Wants fast and reliable network support.
Pain Points	Incomplete requests, unclear forms, and delayed processing.
Needs / Goals	Simple submission, clear tracking, and timely fulfillment.

2.3 Brainstorming & Idea Prioritization

During the ideation phase, different features were identified to improve the network request process.

Idea	Priority
Network Request Catalog Item	High
Catalog Variables	High
Catalog UI Policies	High
Approval Process	High
Flow Designer Automation	High
Network Task Creation	High
Network Team Assignment	High
Request Status Tracking	Medium
Notifications	Medium
Reports and Dashboards	Medium

The high-priority features were selected for the initial implementation because they directly support the core request and fulfillment workflow.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The employee journey begins when a user accesses the Service Portal and ends when the network request is fulfilled.

Stage	Employee Action	System Response
Access	Opens Service Portal	Portal displays available services
Discover	Finds Network Request	Catalog item is displayed
Submit	Enters request details	Variables capture information
Validate	Completes required fields	UI Policies validate the form
Process	Submits request	REQ and RITM are created
Approval	Waits for approval	Approval workflow is triggered
Fulfillment	Request is processed	Network Task is created
Completion	Receives completed service	Request status is updated

3.2 Solution Requirements

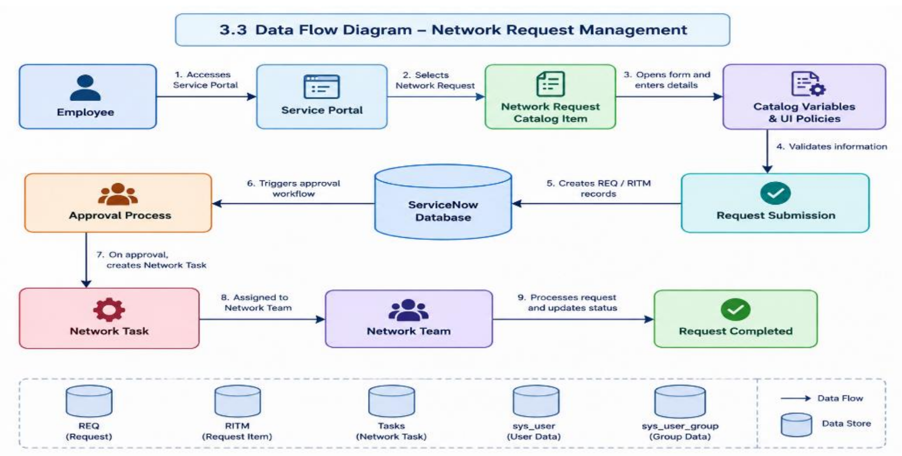
Functional Requirements

ID	Requirement
FR-01	Employees shall access the Network Request Catalog Item.
FR-02	The system shall capture required request information.
FR-03	The system shall validate mandatory fields.
FR-04	Relevant fields shall be displayed dynamically.
FR-05	The system shall create request records after submission.
FR-06	The request shall be sent for approval.
FR-07	A Network Task shall be created after approval.
FR-08	The Network Team shall process the assigned task.

Non-Functional Requirements

ID	Requirement
NFR-01	The system should provide a simple user interface.
NFR-02	The system should use ServiceNow access controls.
NFR-03	The workflow should process requests consistently.
NFR-04	The system should provide normal ServiceNow response performance.

3.3 Logical Data Flow Diagram



3.4 Technology Stack

S.No.	Component	Description	Technology

1	User Interface	Employee-facing request interface	Service Portal / Service Catalog
2	Catalog	Network request submission	Service Catalog
3	Rules	Dynamic fields and validation	Catalog UI Policies
4	Automation	Approval and workflow processing	Flow Designer
5	Data	Request and fulfillment records	REQ / RITM / Tasks
6	References	User and group information	sys_user / sys_user_group

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Parameter	Project Fit
Problem	Employees need a simple and structured way to submit network requests.
Solution	A ServiceNow-based Network Request Management system.
Customer Benefit	Guided request entry and better request visibility.
Operational Benefit	Structured requests improve approvals and fulfillment.
Uniqueness	Combines Service Catalog, UI Policies, approvals, and task automation.
Scalability	Additional request types and automation can be added later.

4.2 Proposed Solution

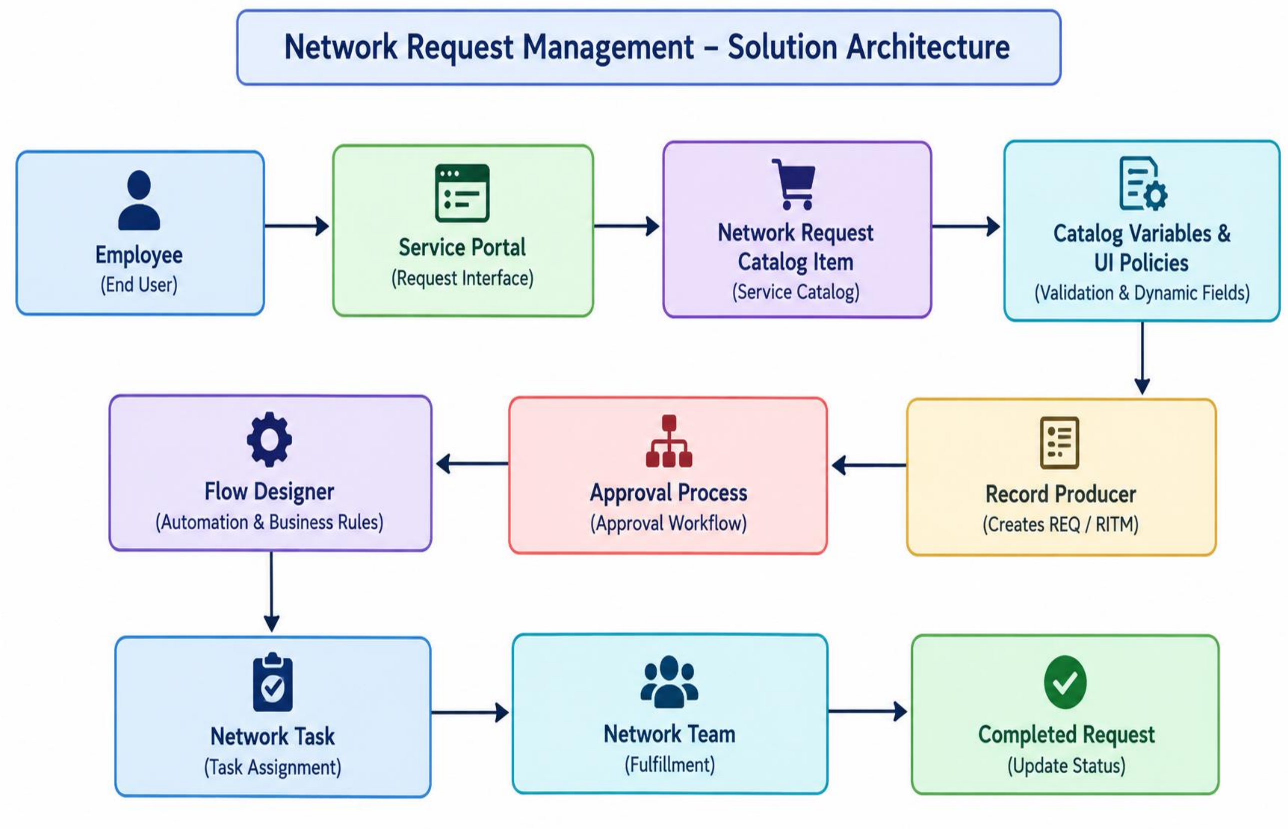
The proposed solution is a **ServiceNow-based Network Request Management system**.

Employees submit network-related requests through a **Network Request Catalog Item**. Catalog variables capture request information, while Catalog UI Policies provide dynamic form behavior and validation.

After submission, the request is processed through an approval workflow. Once approved, a Network Task is created and assigned to the Network Fulfillment Team.

4.3 Solution Architecture

The solution architecture connects the **Service Portal**, **Network Request Catalog**, **validation rules**, **approval process**, and **Network Tasks**.



5.

PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was divided into four implementation sprints.

Sprint	Functional Requirement	User Story / Task	Story Points	Priority	Student Name
Sprint-1	Foundation	Create Network Request components and fields	6	High	Ambica Tenagala
Sprint-2	Catalog & Rules	Create catalog item, variables, and UI Policies	6	High	Ambica Tenagala
Sprint-3	Automation	Configure approvals and Network Task creation	6	High	Ambica Tenagala
Sprint-4	Testing	Testing and documentation	2	High	Ambica Tenagala

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

Test ID	Scenario	Expected Result	Status
TC-01	Open Service Portal	Portal loads successfully	Pass
TC-02	Find Network Request	Catalog item is visible	Pass
TC-03	Open request form	Form opens successfully	Pass
TC-04	Enter request details	Variables are available	Pass
TC-05	Select device type	Relevant options display	Pass
TC-06	Validate fields	Mandatory validation works	Pass
TC-07	Submit request	REQ and RITM are created	Pass
TC-08	Approval process	Request goes for approval	Pass
TC-09	Create Network Task	Task is created	Pass
TC-10	End-to-end test	Complete workflow works	Pass

6.2 User Acceptance Testing (UAT)

UAT ID	Scenario	Acceptance Criteria	Result
UAT-01	Open Service Portal	Portal accessible	Accepted
UAT-02	Find Network Request	Catalog item visible	Accepted
UAT-03	Open form	Request form opens	Accepted
UAT-04	Select Device Type	Option selectable	Accepted
UAT-05	Validate fields	Required validation works	Accepted
UAT-06	Submit request	Request created	Accepted
UAT-07	Approval	Approval works	Accepted
UAT-08	Task creation	Task assigned	Accepted
UAT-09	Complete request	Request completed	Accepted

UAT Conclusion

The tested Network Request Management workflow satisfies the documented acceptance criteria for the implemented scope.

6.3 Performance Testing

The implemented workflow was tested for normal interactive usage within the ServiceNow Personal Developer Instance.

The application successfully handled:

- Catalog form loading.
- Request submission.
- Validation processing.
- Approval workflow execution.
- Network Task creation.

Note: Large-scale concurrent-user load testing was outside the scope of the project.

7. RESULTS

7.1 Output Screenshots

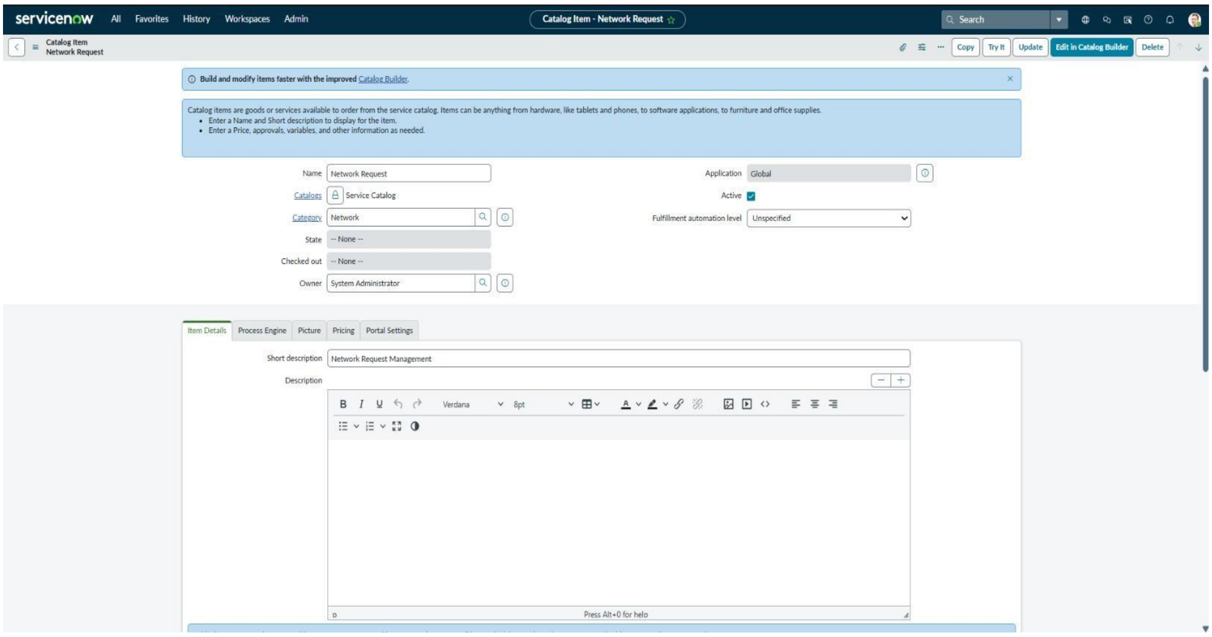


Fig 7.1 Network Request Catalog Item

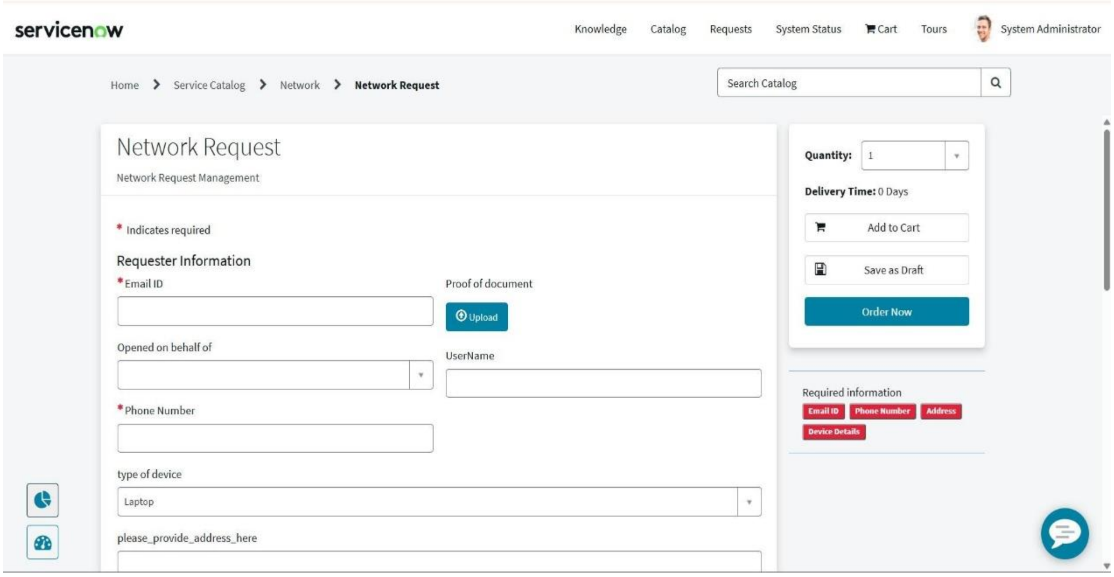


Fig 7.2 Network Request Form

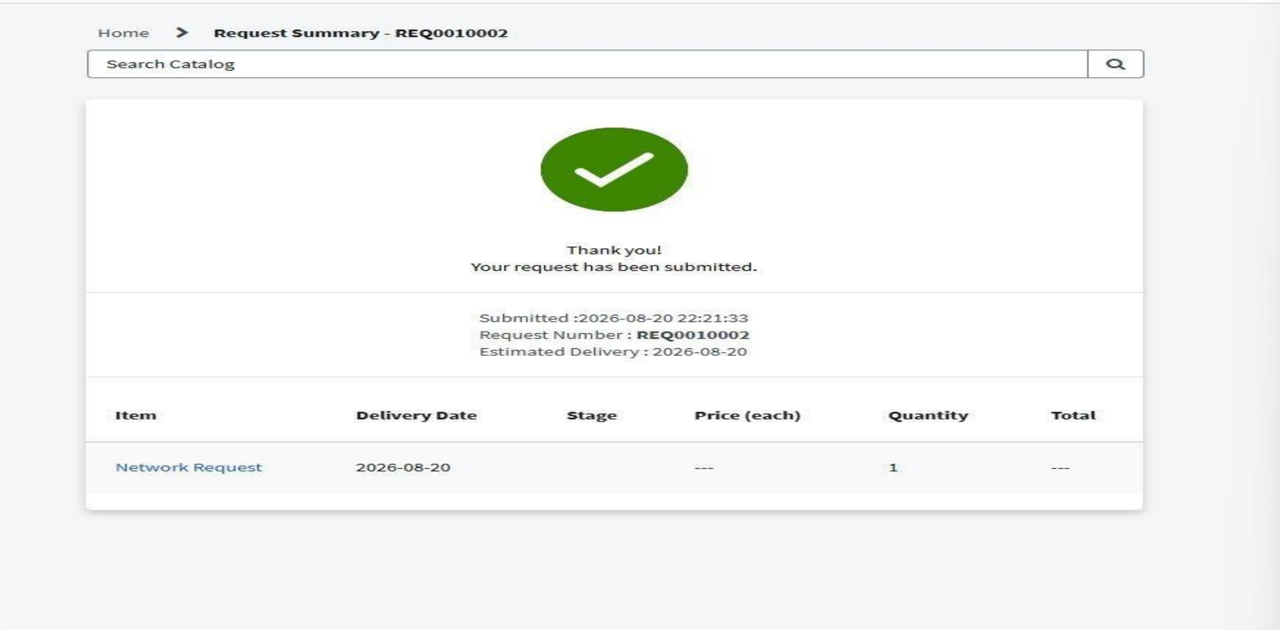


Fig 7.3 Submitted Network Request Record

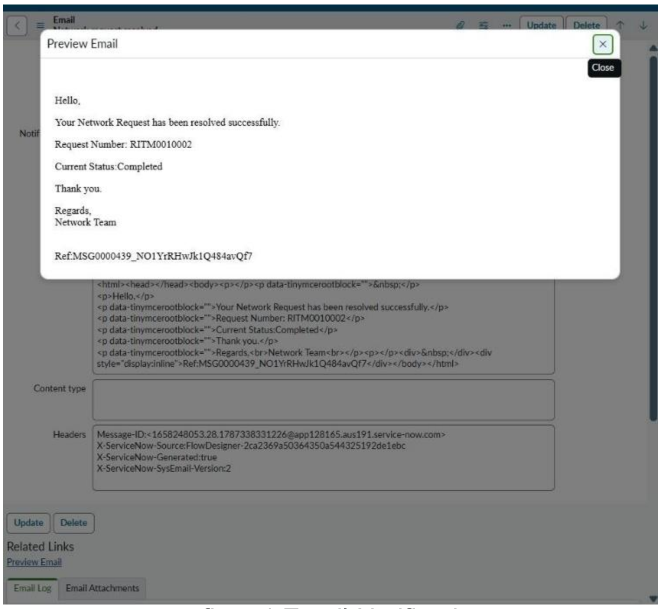


fig 7.4 Email Notification

7.2 Work Item Status Validation

Work Item	Status	Validation
Network Request Catalog	Completed	Catalog item created
Catalog Variables	Completed	Required fields configured
UI Policies	Completed	Dynamic behavior configured
Approval Process	Completed	Approval configured
Flow Designer	Completed	Workflow automation configured
Network Tasks	Completed	Tasks created for fulfillment
Service Portal	Completed	Catalog accessible
Testing	Completed	Workflow validated

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides centralized network request submission.
- Reduces incomplete request information.
- Improves request standardization.
- Automates approvals and task creation.
- Improves visibility into the request lifecycle.

- Reduces manual effort for the Network Team.

Disadvantages

- Depends on correct ServiceNow configuration.
- Initial configuration requires platform knowledge.
- Complex workflows may require additional maintenance.
- Advanced integrations may require additional development.

9. CONCLUSION

The **Automated Network Request & Fulfillment Management System** provides a structured approach to managing network service requests using ServiceNow.

The solution enables employees to submit requests through a centralized catalog, captures required information, validates user input, supports approval processing, and automates Network Task creation.

The project demonstrates how ServiceNow capabilities such as the **Service Catalog, Catalog UI Policies, Flow Designer, Approvals, and Task Management** can be combined to improve operational efficiency and the employee experience.

10. FUTURE SCOPE

The system can be enhanced in the future with:

- Email and mobile notifications.
- Advanced dashboards and reporting.
- SLA monitoring.
- Additional network service categories.
- Automatic priority calculation.
- Integration with external network monitoring tools.
- AI-based request classification.
- Virtual Agent support.
- Knowledge articles for common network issues.

11. APPENDIX

Appendix A – Technology Components

- ServiceNow Service Catalog
- Catalog Items
- Catalog Variables
- Catalog UI Policies
- Flow Designer
- Approval Engine
- REQ and RITM Records
- Network Tasks
- sys_user
- sys_user_group

Appendix B – Project Flow